

Which Structural Prior Belongs in Global Attention?

A Controlled Study of a Hybrid Graph Transformer as a Cable-Dynamics Model

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Abstract

Dynamic models of deformable linear objects are essential components of robotic manipulation frameworks, enabling prediction, planning, and control for objects such as cables, ropes, and sutures. These objects can be represented as graphs, but their dynamics involve two distinct notions of locality: elastic interactions follow arc-length proximity along the object, whereas self-contact is governed by Euclidean proximity. We hold a GPS-style backbone, training, and evaluation windows fixed and change only an additive attention bias: none, Euclidean distance, chain distance, or two allocations of these across heads. Floor-contact rel ℓ_2 is about eight times free-flight rel ℓ_2 ; mixed allocations have the lowest means as a three-seed table order, not as evidence that the useful prior flips with phase.

1 Introduction

Deformable linear objects (DLOs) are slender bodies such as cables, ropes, wires, and tubes, whose length greatly exceeds their cross-section (Caporali et al., 2026). They appear in industrial, domestic, and medical work, including wiring harnesses and surgical sutures (Caporali et al., 2026; Yu et al., 2025a).

A robot that manipulates a DLO must predict how the object will move in response to an actuation or applied force, so those predictions can feed planning and closed-loop control (Yu et al., 2025a; Gu et al., 2025; Yang et al., 2021).

Classical rod models take Cosserat theory as the continuum baseline (Caporali et al., 2026). In manipulation they are often expensive to solve, and their accuracy depends on material and geometric parameters that are hard to obtain for a new cable (Caporali et al., 2026; Yu et al., 2025a). Open-loop execution of a path planned under such a model fails when that identification is wrong (Yu et al., 2025a).

Data-driven models learn DLO behavior from observed or simulated trajectories, with low online cost and little need for an explicit constitutive law (Caporali et al., 2026; Yang et al., 2021; Gu et al., 2025). They can capture nonlinear dynamics that a coarsely identified analytical model misses, but they remain sensitive to the training distribution (Caporali et al., 2026).

Graph-based learned simulators represent the physical state as a graph and roll it forward by message passing. GNS builds a new edge set at each step from particles that fall inside a spatial connectivity radius (Sanchez-Gonzalez et al., 2020). MeshGraphNets instead keeps two edge types: mesh edges for internal dynamics along the discretization, and world edges between nodes that are close in Euclidean space but not already mesh-neighbors, to carry contact and collision (Pfaff et al., 2021). Those choices put locality into the edge set, before the update.

A DLO makes that design load-bearing. It is a one-dimensional chain embedded in three-dimensional space, so it has two notions of locality that need not agree. Elastic interactions follow arc-length proximity along the chain. Self-contact couples segments that are close in Euclidean space even when they are far in chain index. The two neighborhoods agree when the cable is straight and diverge when it folds. MeshGraphNets’ two edge types are exactly those two neighborhoods; a cable is the case where they disagree.

GraphGPS (GPS) changes where that choice can live. A GPS layer runs local message passing on the real edges in parallel with global attention over all node pairs (Rampášek et al., 2022). The two neighborhoods of a cable can therefore stay in the edge set, as they do in MeshGraphNets, or they can be offered to the dense branch as a pairwise term that the original GPS recipe does not include.

We ask whether that dense branch should remain pairwise structure-agnostic, or be biased by

Euclidean distance, chain distance, or a mixture of both. Whether the useful prior depends on which neighborhood dominates is an evaluation question, split by floor contact versus free flight.

We isolate the pairwise metric as an additive attention bias, holding the GPS-style backbone, training, and evaluation windows fixed.

- A matched-budget family of biases: none, Euclidean, chain, and two allocations of these across heads.
- A phase-conditioned evaluation of floor contact versus free flight.

Code, configurations, and the per-run records behind every table cell are at <https://github.com/avihaig/dlogps>.

2 Related Work

Cable dynamics and learned simulators. Classical simulation encodes mechanics explicitly: Discrete Elastic Rods discretizes rod mechanics (Bergou et al., 2008), whereas MuJoCo provides generalized-coordinate dynamics and contact resolution (Todorov et al., 2012). Learned DLO models instead approximate transitions from trajectories, by bidirectional recurrence along the chain or attention over learned interactions (Yang et al., 2021; Gu et al., 2025). GNS and MeshGraphNets already show that proximity edges and split mesh/world edges are useful (Sanchez-Gonzalez et al., 2020; Pfaff et al., 2021), but they commit that choice in the edge set. They do not isolate which metric should guide global attention in cable rollouts.

From local graph attention to hybrid graph transformers. GAT learns weights over adjacent nodes; its attention remains neighborhood-restricted (Veličković et al., 2018). GraphGPS is our backbone (Rampášek et al., 2022). The closest DLO hybrid, EA-PE-GAT, aggregates local interactions before a Transformer encoder for global reasoning in 2D quasi-static shape control (Yu et al., 2025b). It motivates local-global modeling, but its serial architecture and task do not isolate the structural prior inside global attention.

Structural priors inside attention. Dense self-attention does not encode graph locality by itself. Existing methods either restrict which pairs may attend or modify their scores.

These methods inject pairwise structure through additive biases or support masks, mostly on static benchmarks or with several signals coupled (Ying et al., 2021; Luo et al., 2023; Liu et al., 2024; Yun et al., 2026). Our matched-budget ablation holds the GPS backbone and training protocol fixed and assigns global heads no prior, one metric, or a mixture of chain and 3D Euclidean metrics on dynamic cables, where those neighborhoods can disagree.

3 Method

3.1 Cable representation and prediction target

The cable is represented by N vertices joined by $N - 1$ segments (Bergou et al., 2008), with each chain edge stored in both directions. Node features comprise a velocity history, Young’s modulus, joint damping, diameter, effective linear density, and clipped distance to the floor. Edge features contain relative displacement, distance, and rest length. Absolute coordinates are omitted, making the representation invariant to translations parallel to the floor while retaining vertical position through the floor-distance feature.

Young’s modulus, joint damping, diameter, and linear density are mapped by $\log_{10}$ and a fixed affine transform onto $[-1, +1]$: $(\log_{10} x - c)/h$ with (c, h) equal to $(7, 3)$, $(-3, 2)$, $(-2.301, 1)$, and $(-1.3495, 2.6505)$ in that order. Values outside those spans are not clipped. Velocity history, floor distance, edge features, and displacement targets are standardized from the training split, before input noise is added, and those statistics are frozen for training and evaluation. Positions are not scaled. The network predicts per-node displacement; predictions are inverse-standardized and added to the current positions, from which velocities are updated using the recorded frame interval. Training uses one-step targets with random-walk input noise (Sanchez-Gonzalez et al., 2020), whereas evaluation is autoregressive. During rollout, the Euclidean attention bias is computed from the model’s predicted positions rather than ground truth.

3.2 Hybrid local-global architecture

Two linear encoders lift node and edge features to a common width d , the encoded graph passes through L identical layers, and a linear readout maps the final node embeddings to the per-node

variant	$b_h(i, j)$, by head
Unbiased	0 on every head
Euclidean	$\gamma_h \tilde{D}_{ij}$ on every head
Chain	$\beta_h \tilde{C}_{ij}$ on every head
Mixed	$\lfloor H/3 \rfloor$ heads Euclidean, $\lfloor H/3 \rfloor$ chain, the rest unbiased
Euclidean+Chain only	half the heads Euclidean, half chain, none unbiased

Table 1: Attention-bias variants. Mixed and Euclidean+Chain only allocate the Euclidean and chain formulas across heads; neither adds a third distance.

displacement.

Each layer runs a local and a global stream on the *same* input, in parallel rather than in sequence (Rampášek et al., 2022), so message passing cannot smooth the representation before attention sees it. The local stream is a GatedGCN over the chain edges, which reads and updates the edge embeddings no other stream touches. The global stream is dense multi-head attention over all N nodes, seeing node embeddings alone: no edge features, and in the unbiased variant no pairwise term and no mask.

Each stream output is added back to the layer input and is normalized; the two are summed and mixed by a two-layer ReLU MLP of hidden width $2d$, whose output is again added and normalized. Every normalization is post-residual and taken over the feature axis of a single node.

3.3 Attention-bias intervention

For attention head h , the score and normalized attention weight are

$$s_{ij}^h = \frac{q_i^h \cdot k_j^h}{\sqrt{d_k}} - b_h(i, j), \quad (1)$$

$$\alpha_{ij}^h = \frac{\exp(q_i^h \cdot k_j^h / \sqrt{d_k}) \exp[-b_h(i, j)]}{\sum_r \exp(q_i^h \cdot k_r^h / \sqrt{d_k}) \exp[-b_h(i, r)]}. \quad (2)$$

Thus, $b_h(i, j) = 0$ recovers standard attention, while a positive bias attenuates the contribution of node j to node i .

Table 1 lists the five variants. For the Euclidean and chain biases, $\gamma_h, \beta_h = \text{softplus}(\theta_h) > 0$. One rate is learned per biased head, initialized to one and shared across all layers. Euclidean+Chain only isolates the unbiased heads of Mixed: it keeps Mixed’s Euclidean and chain assignments and replaces its unbiased heads with physical priors at the same per-head budget.

Chain and Euclidean distances are normalized as

$$\tilde{C}_{ij} = \frac{|i - j|}{N - 1}, \quad \tilde{D}_{ij} = \frac{\|p_i - p_j\|}{L}, \quad (3)$$

where p_i is the position of node i and L is the cable rest length. The variants differ by at most H learned scalars and are therefore effectively parameter matched.

4 Experiments

4.1 Data generation

We simulate in MuJoCo 3.9.0. Each cable is a 32-segment chain ($N=33$ vertices) under the cable elasticity plugin, with $G/E=0.38$. The integrator is `implicitfast` at $\Delta t=4.2 \times 10^{-4}$ s; kinematics are recorded at 500 Hz. The scene is a floor and two grippers welded to the cable ends. Each episode settles the cable, drives the grippers to a randomized pose, and releases both welds on the same tick; $t=0$ is that instant, after which the cable falls onto the floor.

The population is 46 cables with rest length 0.80 to 1.60 m, diameter 1.5 to 10 mm, and five Young’s moduli from 10^6 to 10^9 Pa. Training uses those cables, 100 episodes each. The test set is 40 cables whose parameters lie inside the same ranges and never appear in training.

4.2 Training

The loss is mean squared error on the standardized one-step displacement, with random-walk input noise $\sigma = 0.1$. Optimization is Adam with gradient clipping at 1.0 and an exponential learning-rate decay from 10^{-3} to 10^{-4} over 100k steps at batch 512. The velocity history is $C = 70$ frames. Weight decay and dropout are zero. Each run trains the full 100k steps; the reported checkpoint is the last.

The backbone uses width $d = 128$, $H = 8$ heads and $L = 4$ layers, 891k parameters in total. Width, noise, history depth, and regularization were chosen on validation windows, not on the test cables.

4.3 Rollouts

Evaluation is autoregressive: each predicted state is fed back as the next input. Euclidean attention bias is recomputed from the predicted positions at every step, as in Section 3. Each rollout has 400 predicted frames; frame zero is the observed input

state. Arms are evaluated on the same windows and joined by (cable, episode, start frame), so no comparison depends on two arms having scored the same windows in the same order.

4.4 Metrics

Let $\hat{p}_i(t)$ and $p_i(t)$ be the predicted and recorded positions of node i at predicted frame t , and let L be that sample's rest length. The per-node Euclidean error and its mean over the N nodes are

$$e_i(t) = \|\hat{p}_i(t) - p_i(t)\|_2. \quad (4)$$

$$\ell_2(t) = \frac{1}{N} \sum_{i=1}^N e_i(t). \quad (5)$$

$$\text{rel } \ell_2(t) = \frac{\ell_2(t)}{L}. \quad (6)$$

For each rollout the reported number is the mean of $\text{rel } \ell_2(t)$ over that rollout's predicted frames.

Link-length drift and self-intersection use the prediction and the cable geometry only, not the recorded trajectory. Recorded rollouts give the simulator's own baseline on both. Rest length per segment, predicted segment length, and per-frame drift are

$$\ell_0 = \frac{L}{N-1}. \quad (7)$$

$$\hat{\ell}_k(t) = \|\hat{p}_{k+1}(t) - \hat{p}_k(t)\|_2. \quad (8)$$

$$\delta(t) = \frac{1}{N-1} \sum_{k=1}^{N-1} \frac{|\hat{\ell}_k(t) - \ell_0|}{\ell_0}. \quad (9)$$

The reported drift is the mean of $\delta(t)$ over the rollout.

Let $d_{jk}(t)$ be the shortest distance between the centerlines of segments j and k , and let D be the cable diameter. A pair with $|j-k| > 2$ (beyond the two-edge bending stencil) violates when $d_{jk}(t) < D$. The reported self-intersection is the mean of

$$v(t) = \mathbf{1} \left[\min_{|j-k| > 2} d_{jk}(t) < D \right] \quad (10)$$

over the rollout.

Table 3 splits local-on $\text{rel } \ell_2$ by phase on the recorded trajectory, never on the prediction. A predicted frame is floor contact when $\min_i z_i \leq D$ after the input frame, and free flight otherwise. Each cell is the mean over selected frames in a rollout, then the equal-weight mean over rollouts with at least one selected frame.

4.5 Comparison design

The main matrix is the five biases of Table 1, each at three seeds, with the local stream on and with it off.

Local on keeps both streams of Section 3: GatedGCN over the chain edges, and dense global attention with that variant's bias. Local off drops the GatedGCN module, so only global attention remains. Chain only drops the attention module instead: message passing on the chain edges, no pairwise bias.

Two models sit outside the GPS block. The per-node MLP maps each node's own history to that node's displacement and never mixes nodes. The global LSTM reads the flattened N -node state as a C -step sequence, plus centroid-relative positions divided by rest length, and writes all N displacements. The MLP does not receive those positions: a centroid would mix nodes.

Each seed is reduced to its mean over evaluation windows, together with that seed's window standard deviation. The reported triple is the mean of the three seed means, the mean of the three window standard deviations, and the standard deviation of the three seed means. Table 2 marks the lowest mean in each local-on or local-off block. Table 3 marks the lowest mean in each phase column. No significance bar is applied.

5 Results

Table 2 reports $\text{rel } \ell_2$, link-length drift δ , and self-intersection v on the unseen-cable test set.

A physical attention bias lowers mean $\text{rel } \ell_2$ relative to unbiased attention. With the local stream on, Euclidean+Chain only has the lowest means on $\text{rel } \ell_2$ and v , and Mixed has the lowest mean drift. With the local stream off, Chain has the lowest mean $\text{rel } \ell_2$, Mixed the lowest mean drift, and Euclidean+Chain only the lowest mean v . The local-off gaps are larger: Chain is 15% below unbiased on mean $\text{rel } \ell_2$, against 9% for Euclidean+Chain only versus unbiased with local on. The local-on ranking is a three-seed table order, not a separated effect.

GPS rows have lower mean $\text{rel } \ell_2$ than chain-only message passing. Chain with local off is 63% below chain-only on mean $\text{rel } \ell_2$. Euclidean+Chain only with local on is 68% below. Dense attention already has lower mean error than chain-only message passing, and the local stream plus a mixed bias has lower mean error still. Local-

	$\text{rel } \ell_2$ (frac. of L)	δ (frac. of ℓ_0)	v (frac. of frames)
<i>local stream on</i>			
Unbiased	$0.0325 \pm 0.0281 \pm 0.0058$	$0.0300 \pm 0.0159 \pm 0.0032$	$0.171 \pm 0.231 \pm 0.009$
Euclidean	$0.0320 \pm 0.0267 \pm 0.0042$	$0.0303 \pm 0.0135 \pm 0.0021$	$0.174 \pm 0.229 \pm 0.032$
Chain	$0.0316 \pm 0.0341 \pm 0.0057$	$0.0271 \pm 0.0124 \pm 0.0023$	$0.167 \pm 0.227 \pm 0.030$
Mixed	$0.0302 \pm 0.0251 \pm 0.0067$	$0.0267 \pm 0.0109 \pm 0.0030$	$0.168 \pm 0.229 \pm 0.012$
Euclidean+Chain only	$0.0297 \pm 0.0288 \pm 0.0035$	$0.0275 \pm 0.0129 \pm 0.0007$	$0.165 \pm 0.229 \pm 0.012$
<i>local stream off</i>			
Unbiased	$0.0399 \pm 0.0246 \pm 0.0002$	$0.1693 \pm 0.1891 \pm 0.0030$	$0.216 \pm 0.258 \pm 0.006$
Euclidean	$0.0382 \pm 0.0279 \pm 0.0007$	$0.1470 \pm 0.2232 \pm 0.0178$	$0.208 \pm 0.262 \pm 0.020$
Chain	$0.0339 \pm 0.0195 \pm 0.0008$	$0.0783 \pm 0.0382 \pm 0.0040$	$0.174 \pm 0.241 \pm 0.004$
Mixed	$0.0352 \pm 0.0211 \pm 0.0016$	$0.0779 \pm 0.0455 \pm 0.0036$	$0.172 \pm 0.242 \pm 0.016$
Euclidean+Chain only	$0.0343 \pm 0.0202 \pm 0.0002$	$0.0784 \pm 0.0409 \pm 0.0013$	$0.161 \pm 0.234 \pm 0.006$
<i>baselines</i>			
chain only	$0.0925 \pm 0.0838 \pm 0.0720$	$0.0812 \pm 0.0589 \pm 0.0322$	$0.115 \pm 0.191 \pm 0.024$
MLP	$0.0868 \pm 0.0785 \pm 0.0163$	$1.35 \pm 2.32 \pm 0.73$	$0.435 \pm 0.256 \pm 0.010$
LSTM	$0.1070 \pm 0.0911 \pm 0.0397$	$0.2819 \pm 0.3124 \pm 0.0509$	$0.082 \pm 0.150 \pm 0.031$

Table 2: Unseen-cable test, final checkpoint. Variant names are those of Table 1. Cell format: mean of seed means $\pm$ mean of window sds $\pm$ sd of seed means. Bold: lowest mean in each local-on or local-off block. Encoding as in Section 3. 400 windows of 400-frame rollouts.

	floor contact $\text{rel } \ell_2$ (frac. of L)	free flight $\text{rel } \ell_2$ (frac. of L)
Unbiased	$0.0511 \pm 0.0536 \pm 0.0105$	$0.0062 \pm 0.0095 \pm 0.0001$
Euclidean	$0.0510 \pm 0.0615 \pm 0.0080$	$0.0062 \pm 0.0099 \pm 0.0005$
Chain	$0.0529 \pm 0.0871 \pm 0.0122$	$0.0064 \pm 0.0131 \pm 0.0014$
Mixed	$0.0484 \pm 0.0551 \pm 0.0135$	$0.0058 \pm 0.0097 \pm 0.0011$
Euclidean+Chain only	$0.0479 \pm 0.0643 \pm 0.0068$	$0.0061 \pm 0.0127 \pm 0.0010$

Table 3: Local-on unseen-cable test, final checkpoint. Phase labels follow Section 4. Cell format matches Table 2. Bold: lowest mean in the column. Eligible windows: 400 floor contact, 345 free flight.

on rows have lower mean drift: mean drift stays near 0.03 with local on and rises to 0.078–0.169 with local off.

The per-node MLP and the global LSTM do not match the GPS rows. The MLP mean drift is 1.35. The LSTM mean self-intersection is the lowest in the table, but mean drift is 0.2819 and mean $\text{rel } \ell_2$ is 0.1070.

Table 3 splits the local-on rows of Table 2 by phase.

Floor-contact $\text{rel } \ell_2$ is about eight times free-flight $\text{rel } \ell_2$ on every local-on row, and about two thirds of predicted frames are floor contact. Among the five variants, Euclidean+Chain only has the lowest floor-contact mean and Mixed the lowest free-flight mean. Euclidean is near unbiased, and Chain is highest in both columns. That column order is a three-seed table order, not a claim that the useful prior flips with phase.

6 Conclusion

A GPS-style block with both streams, plus Euclidean and chain biases allocated across heads (Euclidean+Chain only), has the lowest mean rollout error on this cable set with the local stream on. With the local stream off the gaps are larger, and Chain has the lowest mean $\text{rel } \ell_2$. GPS rows have lower mean error than chain-only message passing; local-on rows have lower mean drift. The structure-free MLP and LSTM do not.

Floor-contact error is about eight times free-flight error; mixed allocations have the lowest phase means as a three-seed table order, not as evidence that the useful prior flips with phase.

The local-on ranking among biases is a table order, not a separated effect. The numbers are this MuJoCo population at $N = 33$. Euclidean node distance is a proxy for segment contact. Training is one-step; evaluation is autoregressive. Mixed heads with the lowest means do not by themselves show head specialization.

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